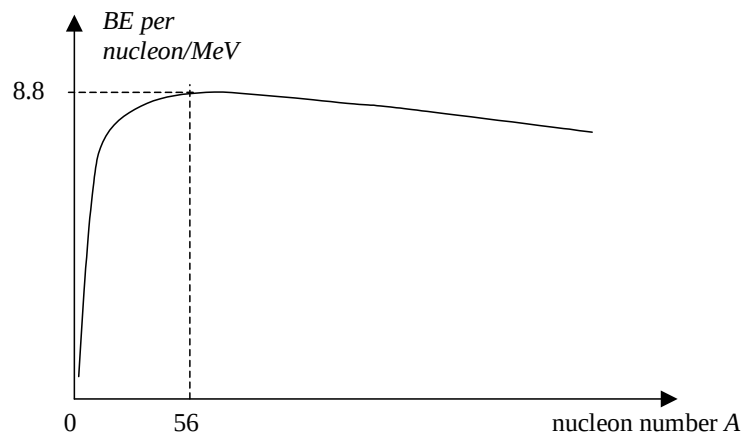


- 6 (a) The binding energy of a nucleus is the energy released when a nucleus is formed from its constituent protons and neutrons. B1

(b)(i)



Correct Shape
Correct labelling for max BE B1

- (b)(ii) In nuclear fission, a heavy nucleus splits to give two daughter nuclei of greater binding energy per nucleon. Hence, there is an overall energy release during the process. B1

- (c)(i) A **chain reaction** is able to occur as the neutrons released in the fission produce an additional fission in at least one further nucleus. B1

- (c)(ii) Energy released in one reaction = $(m_u + m_n - (m_{xe} + m_{sr} + 2m_n)) c^2$ M2
 $= (235.043929 + 1.008665 - (138.918793 + 94.919359 + 2(1.008665))) \times$
 $\times [3 \times 10^8]^2$ A1
 $= 2.94 \times 10^{-11} \text{ J}$

- (c)(iii) No of reactions that will take place in 2 kg Uranium M1
 $= (2000/235) \times 6.02 \times 10^{23}$
 $= 5.12 \times 10^{24}$
 Energy from 2 kg of Uranium = $2.94 \times 10^{-11} \times 5.12 \times 10^{24}$ M1
 Useful Energy from 2 kg of Uranium = $0.080 \times 2.94 \times 10^{-11} \times 5.12 \times 10^{24}$ M1
 $= 1.21 \times 10^{13} \text{ J}$
 Amount of time $t = (1.21 \times 10^{13} / (50 \times 10^{12})) \times 12 = 2.9 \text{ months}$ A1

7 (a) Random – it cannot predict which and when a nucleus will decay R1